

**SALES ORBIS: AI-BASED SALES PIPELINE INTELLIGENCE
AND DEAL WIN PREDICTION PLATFORM**

PHASE I REPORT

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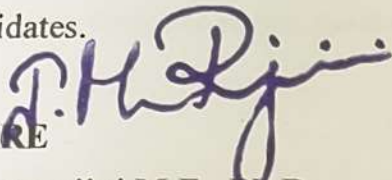
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BONAFIDE CERTIFICATE

Certified that this Report titled "SALES ORBIS: AI-BASED SALES PIPELINE INTELLIGENCE AND DEAL WIN PREDICTION PLATFORM" is the bonafide work of **DHIVYA SHREE K (230701079)**, **HARINI M (230701101)**, **DHARANIKUMAR S (230701073)**, **DEEPAK R (230701067)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

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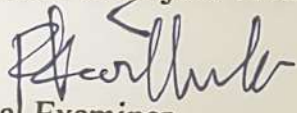
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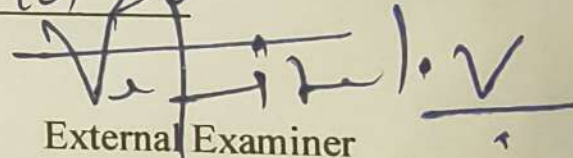
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ABSTRACT

Over multi-stage projects, multiple suppliers, B2B opportunities, quotations, evaluations, activities, pipeline stages management and domain process data, B2B procurement and sales pipeline management is complicated and data intensiveness. Procurement team needs to understand supplier, commercial, evaluation, engagement, competition and project data to make reliable award decision. Existing procurement and customer relationship management platforms mainly provide descriptive information and opportunity prioritization, supplier evaluation and prediction results depend heavily on manual analysis, history knowledge and experience. This project, the Sales Orbis: AI-Enabled Sales Pipeline Intelligence and Deal Win Prediction Platform, is a full-stack ML/AI driven procurement and supplier data-to-Pipeline intelligence and decision support platform.

This ML platform adopts a dynamic pipeline snapshot-centric technique, that the information at a certain as of date can be used to predict a likelihood that the supplier will end up with the award, by eliminating information leakage from future stages and outcomes. This project features a total of 11 structured procurement and supplier data sources containing information about projects, suppliers, opportunities, quotations, evaluations, activities, stage history, project status and outcomes, as well as a leakage-safe Gold ML feature table with 34999 prediction snapshots for 24857 opportunities. The machine learning pipeline includes data validation, data pre-

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CHAPTER 6 CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

This project phase aimed to build an AI-driven sales pipeline intelligence platform to help predict deal-win probabilities and facilitate data-driven procurement and sales forecasting analysis. We operationalized the entire prediction lifecycle, whereby procurement data is validated, sanitised and made leakage-safe features, opportunities are modelled as snapshots over time, multiple ML models are trained and evaluated, and a best performer is used to generate deal-win probabilities, deal outcome predictions, risk analysis, and expected procurement values. The pipeline's outputs were then visualised in a BI report that is standalone to estimate deal pipeline health, compute open and closed deals, produce individual deal insights, and project future sales pipeline. We answer all the key requirements for dynamic deal-win prediction. Leakage-safe feature engineering is performed based on the information available at each asofdate, and the project-based and temporal data splits make sure that no project overlap is allowed and no future information is used in evaluating the models. Logistic Regression and Decision Tree baselines are tested against more advanced approaches including Random Forest, XGBoost, LightGBM and Cat Boost models. The models are evaluated according to ROC-AUC, PR-AUC, calibrated log-loss, Brier score and temporal robustness. The final model is also inspected in terms of probability calibration and between-training-vs.-test generalisation. It can also give model transparency via SHAP based global and local explanations and diagnosis of prediction behavior at the stage of each pipeline step and each snapshot. Its final prediction module outputs a prediction for each Gold snapshot, and the business report transforms all into actionable sales intelligence, including predicted award rate, risk degree, active-deal details, and forecast value. This bridges the gap between the machine learning pipeline and real-world business